SPECIAL CONTINUOUS RANDOM VARIABLES

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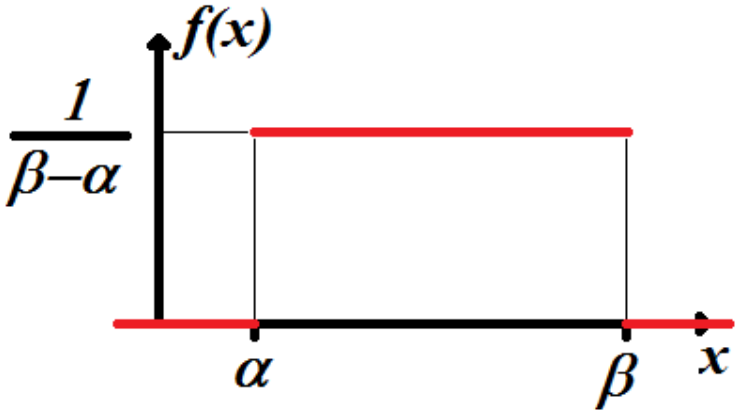
Special Continuous Random Variables

- In these section, we are going to learn about some common types of continuous random variables.
- Many measurable random values in real-life can be modeled with the following random variables.
  - Continuous Uniform Random Variable
  - Exponential Random Variable
  - Normal Random Variable
  - ...
- Due to their special nature, these random variables can be classified according to their PDFs.

Continuous Uniform RV: PDF

- A continuous random variable is said to be **uniformly distributed** over the interval (α, β) if its PDF is given by:

$$f(x) = \begin{cases} \frac{1}{\beta-\alpha}, & \alpha < x < \beta \\ 0, & \text{elsewhere} \end{cases}$$



- The above function is a PDF, since:

$$\int_{\alpha}^{\beta} \frac{1}{\beta-\alpha} dx = \frac{x}{\beta-\alpha}\Big|_{\alpha}^{\beta} = \frac{\beta-\alpha}{\beta-\alpha} = 1$$

since the area under the function is always 1.

Continuous Uniform RV: PDF (cont'd)

- Notice that, the PDF of the **continuous uniform RV** is a constant function.
  - The RV is equally likely to get any real-value between α and β.
- Here, α and β are the **lower bound** and the **upper bound parameters** of the continuous uniform distribution.
  - For different values of α and β such that $\alpha < \beta$, we get a different uniform random variable.
  - If we choose $\alpha = 0$ and $\beta = 1$, the resulting random variable is called the "standard uniform RV".

Continuous Uniform RV: PDF (cont'd)

- If X is a continuous uniform RV with lower bound α and upper bound β:

$$X \sim Unif(\alpha, \beta)$$

then given $a, b \in \mathbb{R}$ such that $\alpha \leq a < b \leq \beta$, the probability that X to take values in interval (a, b) is:

$$P\{a < X < b\} = \int_a^b \frac{1}{\beta - \alpha} dx = \frac{x}{\beta - \alpha}\Big|_a^b = \frac{b-a}{\beta - \alpha}$$

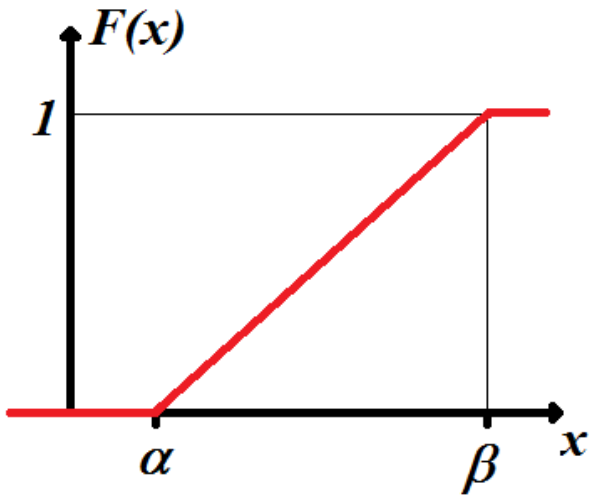
Continuous Uniform RV: CDF

- Suppose $X \sim Unif(\alpha, \beta)$. For $x \in (\alpha, \beta)$, the CDF of the continuous uniform RV is calculated as:

$$F(x) = P\{X \leq x\} = \int_{\alpha}^{x} \frac{1}{\beta-\alpha} dt = \frac{t}{\beta-\alpha}\Big|_{\alpha}^{x} = \frac{x-\alpha}{\beta-\alpha}$$

Thus, we can write the CDF in the following form:

$$F(x) = \begin{cases} 0, & x \leq \alpha \\ \frac{x-\alpha}{\beta-\alpha}, & \alpha < x < \beta \\ 1, & x \geq \beta \end{cases}$$



Continuous Uniform RV: Example

Q1. Buses arrive at a specified stop at 15-minute intervals starting at 7 AM. In other words, they arrive at 7:00, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7:00 and 7:30, find the probability that s/he waits

a. less than 5 minutes for a bus;
b. more than 7 minutes for a bus.

A1. The time is a continuous phenomenon. Thus the question tells us that the arrival time of the passenger is:

$$X \sim Unif\,(\alpha = 0, \beta = 30)$$

if we start the day at 7:00 AM.

Continuous Uniform RV: Example (cont'd)

a. The passenger will have to wait less than 5 minutes if s/he arrives between 7:10 and 7:15 or between 7:25 and 7:30. Hence the desired probability is:

$$\begin{aligned} P\{10 < X < 15\} + P\{25 < X < 30\} &= \int_{10}^{15} \frac{1}{30} dx + \int_{25}^{30} \frac{1}{30} dx \\ &= \frac{5}{30} + \frac{5}{30} = \frac{1}{3} \end{aligned}$$

b. The passenger will have to wait more than 7 minutes if s/he arrives between 7:00 and 7:08 or between 7:15 and 7:23. Hence the desired probability is:

$$\begin{aligned} P\{0 < X < 8\} + P\{15 < X < 23\} &= \int_{0}^{8} \frac{1}{30} dx + \int_{15}^{23} \frac{1}{30} dx \\ &= \frac{8}{30} + \frac{8}{30} = \frac{8}{15} \end{aligned}$$

Continuous Uniform RV: Expected Value

- Suppose $X \sim Unif(\alpha, \beta)$. Then the **expected value** of the continuous uniform RV is:

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f(x)\, dx = \int_{\alpha}^{\beta} \frac{x}{\beta-\alpha} dx = \left. \frac{x^2}{2(\beta-\alpha)} \right|_{\alpha}^{\beta} \\ &= \frac{\beta^2-\alpha^2}{2(\beta-\alpha)} = \frac{(\beta-\alpha)(\beta+\alpha)}{2(\beta-\alpha)} = \frac{\alpha+\beta}{2} \end{aligned}$$

Continuous Uniform RV: Variance

- Suppose $X \sim Unif(\alpha, \beta)$. In order to calculate the variance of the continuous uniform RV, we first need to calculate $E\left[X^2\right]$:

$$\begin{aligned} E\left[X^2\right] &= \int_{-\infty}^{\infty} x^2 f(x)\, dx = \int_{\alpha}^{\beta} \frac{x^2}{\beta-\alpha} dx = \left.\frac{x^3}{3(\beta-\alpha)}\right|_{\alpha}^{\beta} \\ &= \frac{\beta^3-\alpha^3}{3(\beta-\alpha)} = \frac{(\beta-\alpha)\left(\alpha^2+\alpha\beta+\beta^2\right)}{3(\beta-\alpha)} = \frac{\alpha^2+\alpha\beta+\beta^2}{3} \end{aligned}$$

- Then the **variance** of the continuous uniform RV is:

$$\begin{aligned} Var[X] &= \frac{\alpha^2+\alpha\beta+\beta^2}{3} - \left(\frac{\alpha+\beta}{2}\right)^2 = \frac{\alpha^2+\alpha\beta+\beta^2}{3} - \frac{\alpha^2+\beta^2+2\alpha\beta}{4} \\ &= \frac{4\alpha^2+4\alpha\beta+4\beta^2}{12} - \frac{3\alpha^2+3\beta^2+6\alpha\beta}{12} = \frac{\alpha^2-2\alpha\beta+\beta^2}{12} = \frac{(\beta-\alpha)^2}{12} \end{aligned}$$

Continuous Uniform RV: Example II

Q2. Suppose the daily demand to milk in a dairy farm is uniformly distributed between 50 litres and 150 litres.

a. What is the probability that the daily demand is between 80 and 120 litres?
b. Calculate the expected daily demand.
c. Calculate the variance of the daily demand.
d. It is known that today's milk demand was no more than 110 litres, what is the probability that it was more than 85 litres?

A2. The demand to milk can get continuous values. Thus, the question gives us only the following information:

$$X \sim Unif\,(\alpha = 50, \beta = 150)$$

Continuous Uniform RV: Example II (cont'd)

a. Either integrate the PDF:

$$P\{80 \leq X \leq 120\} = \int_{80}^{120} \frac{1}{150-50} dx = \frac{x}{100}\Big|_{80}^{120} = 0.4$$

or use directly the CDF:

$$P\{80 \leq X \leq 120\} = F(120) - F(80) = \frac{120-50}{150-50} - \frac{80-50}{150-50} = 0.4$$

b. $E[X] = \frac{50+150}{2} = \frac{200}{2} = 100$

c. $Var[X] = \frac{(150-50)^2}{12} = \frac{10000}{12} \approx 833.3333$

Continuous Uniform RV: Example II (cont'd)

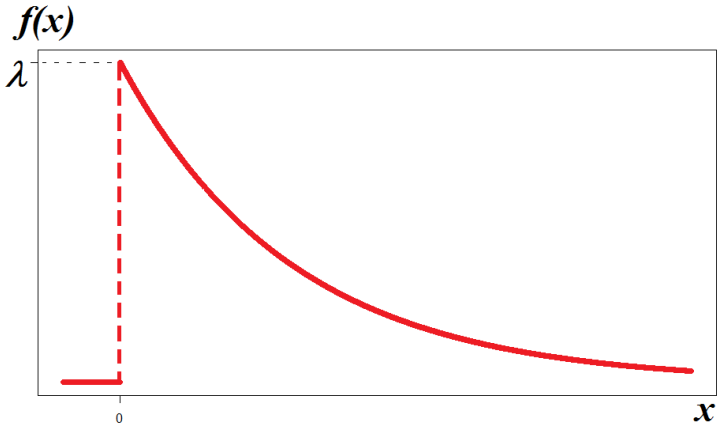
d. Here, the question asks a conditional probability:

$$
\begin{aligned}
P\{X > 85 | X < 110\} &= \frac{P\{X>85 \cap X<110\}}{P\{X<110\}} \\
&= \frac{P\{85<X<110\}}{P\{X<110\}} \\
&= \frac{F(110)-F(85)}{F(110)} \\
&= \frac{\frac{110-50}{150-50} - \frac{85-50}{150-50}}{\frac{110-50}{150-50}} \\
&= \frac{\frac{60}{100} - \frac{35}{100}}{\frac{60}{100}} = \frac{5}{12}
\end{aligned}
$$

Exponential RV: PDF

- A continuous random variable is said to be **exponentially distributed**, if, for $\lambda > 0$, its PDF is given by:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$



- The above function is a PDF, since:

$$\int_0^\infty \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x}\right]_0^\infty = \left[-e^{-\lambda(\infty)}\right] - \left[-e^0\right] = 0 - (-1) = 1$$

since the area under the function is always 1.

Exponential RV: PDF (cont'd)

- Notice that, the PDF of the **exponential RV** is a decaying function (converging to zero).
- Here, λ is the **rate parameter** of the exponential distribution.
  - For different values of $\lambda > 0$, we get a different exponential random variable.

$$X \sim Exp(\lambda)$$

  - If we choose $\lambda = 1$, the resulting random variable is called the "standard exponential RV".
- An exponential RV can get non-negative real values ($X \in [0, \infty)$).

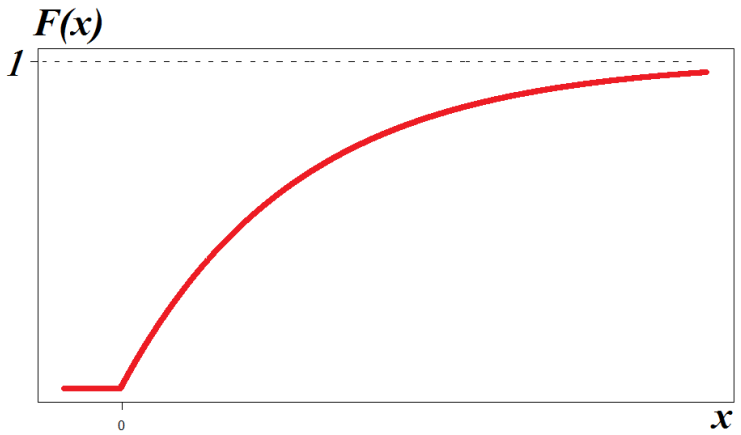
Exponential RV: CDF

- Suppose $X \sim Exp(\lambda)$. For $x \in [0, \infty)$, the CDF of the exponential RV is calculated as:

$$F(x) = P\{X \leq x\} \quad = \int_0^x \lambda e^{-\lambda t} dt = \left[-e^{-\lambda t}\right]_0^x$$
$$= \left[-e^{-\lambda x}\right] - \left[-e^0\right] = 1 - e^{-\lambda x}$$

Thus, we can write the CDF in the following form:

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$$



Exponential RV: Example

Q3. Suppose that the length of a phone call **in minutes** is an exponential random variable with rate parameter $\lambda = 1/10$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait:

a. more than 10 **minutes.**
b. between 10 and 20 **minutes.**

A3. The question tells us that the waiting time, X, is an exponential random variable.

$$X \sim Exp\left(\lambda = \frac{1}{10}\right)$$

Note that, the questions ask probabilities of events given in minutes, which is also the time unit for X.

Exponential RV: Example (cont'd)

a. Either integrate the PDF:

$$P\{X > 10\} = \int_{10}^{\infty} \frac{1}{10} e^{-\frac{x}{10}} = -e^{-\frac{x}{10}}\Big|_{10}^{\infty} = 0 - \left(-e^{-\frac{10}{10}}\right) = e^{-1} \approx 0.3679$$

or use directly the CDF:

$$P\{X > 10\} = 1 - P\{X \leq 10\} = 1 - F(10) = 1 - \left(1 - e^{-\frac{10}{10}}\right) = e^{-1} \approx 0.3679$$

b. Let us use the CDF, directly:

$$\begin{aligned} P\{10 \leq X \leq 20\} &= P\{X \leq 20\} - P\{X < 10\} \\ &= F(20) - F(10) \\ &= \left(1 - e^{-\frac{20}{10}}\right) - \left(1 - e^{-\frac{10}{10}}\right) \\ &= e^{-1} - e^{-2} \approx 0.3679 - 0.1353 = 0.2326 \end{aligned}$$

Exponential RV: Example II

Q4. In a bin, there are 8 batteries, 3 of them being of brand A and 5 of them being of brand B. Though the brands are different, the batteries are indistinguishable. It is known that the lifetime of a brand A battery is uniformly distributed between 30 and 70 hours, whereas the hourly lifetime of a brand B battery is exponentially distributed with rate parameter $\lambda = 0.02$. What is the probability for a randomly selected battery from the bin to work for at most 45 hours?

A4. Let X_A and X_B denote the hourly working lifetime of batteries of, respectively, brand A and brand B.

$$X_A \sim Unif\left(\alpha = 30, \beta = 70\right), \quad X_B \sim Exp\left(\lambda = 0.02\right)$$

Exponential RV: Example II (cont'd)

A4. Let X be the hourly working lifetime of the randomly selected battery. The question asks:

$$P\{X \leq 45\}$$

We only know the distribution of X conditioned on that we know the brand of the battery we select. Hence, we will condition it on the brand. Let:

- A: The event of selecting a type A battery, $P(A) = \frac{3}{8}$.
- A^c: The event of selecting a type B battery, $P(A^c) = \frac{5}{8}$.

Then:

$$\begin{aligned} P\{X \leq 45\} &= P(A)P\{X \leq 45|A\} + P(A^c)P\{X \leq 45|A^c\} \\ &= P(A)P\{X_A \leq 45\} + P(A^c)P\{X_B \leq 45\} \\ &= \frac{3}{8}\left(\frac{45-30}{70-30}\right) + \frac{5}{8}\left(1 - e^{-0.02\times 45}\right) \\ &\approx 0.1406 + 0.3709 = 0.5115 \end{aligned}$$

Exponential RV: Expected Value

- Suppose $X \sim Exp(\lambda)$. Then the **expected value** of the exponential RV is

$$E[X] = \int_0^\infty x f(x)\, dx = \int_0^\infty x\lambda e^{-\lambda x} dx$$

which can be solved through integration by parts. We consider:

$$\int u dv = uv - \int v du$$

where:

$$\begin{aligned} u &= x, & du &= dx, \\ dv &= \lambda e^{-\lambda x} dx, & v &= -e^{-\lambda x} \end{aligned}$$

Therefore:

$$\begin{aligned} E[X] &= \int_0^\infty x\lambda e^{-\lambda x} dx = \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x} dx \\ &= (0-0) + \left[-\tfrac{1}{\lambda} e^{-\lambda x}\right]_0^\infty = 0 - \left(-\tfrac{1}{\lambda}\right) = \tfrac{1}{\lambda} \end{aligned}$$

Exponential RV: Variance

- Suppose $X \sim Exp(\lambda)$. In order to calculate the variance of the exponential RV, we first need to calculate $E\left[X^2\right]$, again through integrating by parts:

$$E\left[X^2\right] = \int_0^\infty x^2 f(x)\,dx = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$$

$$\begin{aligned} u &= x^2, & du &= 2x dx, \\ dv &= \lambda e^{-\lambda x} dx, & v &= -e^{-\lambda x} \end{aligned}$$

$$\begin{aligned} E\left[X^2\right] &= \int_0^\infty x^2 \lambda e^{-\lambda x} dx \\ &= \left[-x^2 e^{-\lambda x}\right]_0^\infty + 2\int_0^\infty x e^{-\lambda x} dx \qquad \Leftarrow \int_0^\infty x e^{-\lambda x} dx = \frac{E[X]}{\lambda} = \frac{1}{\lambda^2} \\ &= (0-0) + \frac{2}{\lambda^2} = \frac{2}{\lambda^2} \end{aligned}$$

Exponential RV: Variance (cont'd)

- Then the **variance** of the exponential RV is:

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Exponential RV: Example III

Q5. The light bulbs in a production lot has lifetimes that follow exponential distribution with mean value 5000 hours.

a. Which value exceeds the lifetimes of the half of the light bulbs?
b. Find the lifetime that is exceeded with probability 0.05.

A5. Let X denote the lifetime of a light bulb:

$$E[X] = 5000\, hours \Rightarrow X \sim Exp\left(\lambda = 0.0002\, \frac{\text{bulb}}{\text{hour}}\right)$$

Part (a) asks the critical value x_1, for which:

$$P\{X \leq x_1\} = 0.5$$

whereas part (b) asks the critical value x_2, for which:

$$P\{X > x_2\} = 0.05 \Rightarrow P\{X \leq x_2\} = 0.95$$

Exponential RV: Example III (cont'd)

A5. Note that:

$$P\{X \leq x\} = F(x) = 1 - e^{-\lambda x} = p \quad \Rightarrow x = -\frac{\ln(1-p)}{\lambda}$$

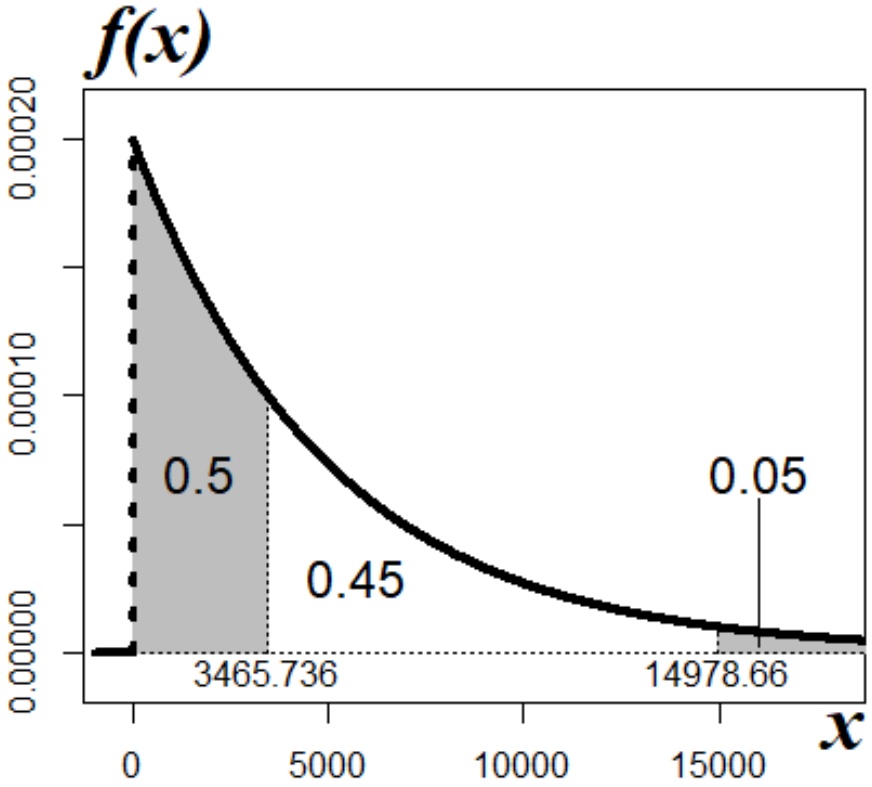
a. $P\{X \leq x_1\} = F(x_1) = 0.5$

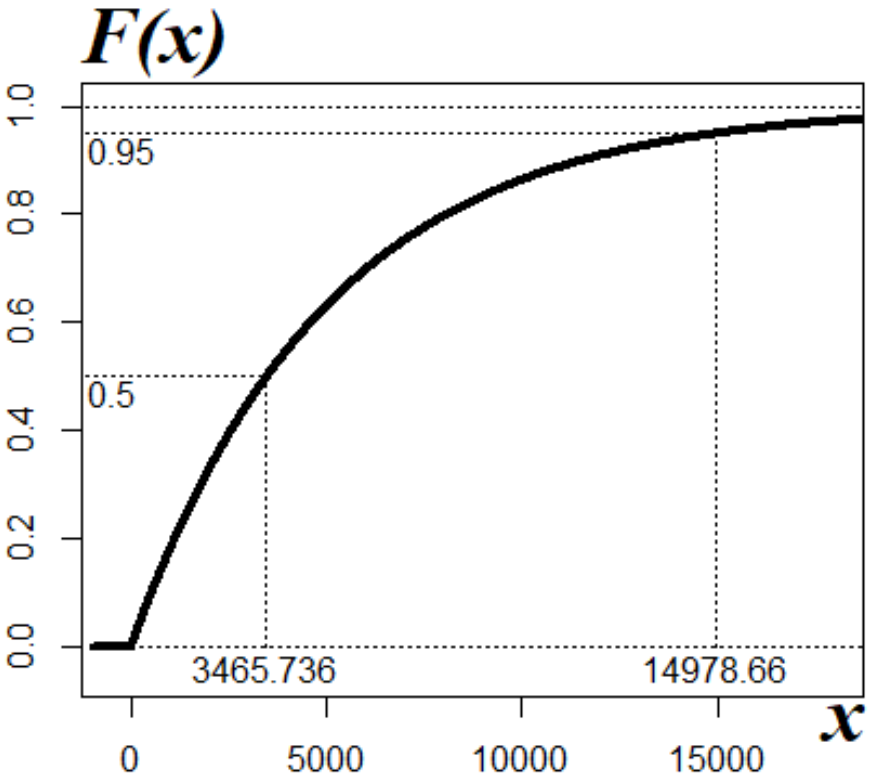
$$x_1 = -\frac{\ln(1-0.5)}{0.0002} \approx 3465.736 \text{ hours}$$

b. $P\{X \leq x_2\} = F(x_2) = 0.95$

$$x_2 = -\frac{\ln(1-0.95)}{0.0002} \approx 14978.66 \text{ hours}$$

Exponential RV: Example III (cont'd)





The Origin of Exponential Random Variable

- Consider a certain event such as arrival of customers to a supermarket.
- Suppose the number of arrivals that occur in an hour follows Poisson distribution with rate parameter λ.
  - On the average λ arrivals occur per hour.
- In that case, the time between two consecutive arrivals is exponentially distributed again with rate parameter λ.
  - The average time between two consecutive arrivals is $\frac{1}{\lambda}$ hours.
- While the Poisson RV models the frequency of events, exponential RV models the period[1] of these events.

[1] Period: Time between consecutive events.

Exponential RV: Example IV

Q6. In a certain district, the number of traffic accidents occuring in any month follows Poisson distribution with a mean of 4 accidents per month.

a. What is the expected time between two consecutive accidents?
b. What is the variance of the time between two consecutive accidents?
c. What is the probability that the time between two consecutive accidents to be more than six weeks?

Exponential RV: Example IV (cont'd)

A6. Though the question mentions the Poisson distribution, the time between Poisson events are exponentially distributed.

$$X \sim Exp\left(\lambda = 4\frac{\text{accident}}{\text{month}}\right)$$

a. $E[X] = \frac{1}{\lambda} = \frac{1}{4} = 0.25 \text{ month}$

b. $Var[X] = \frac{1}{\lambda^2} = \frac{1}{4^2} = 0.0625$

c. Six weeks is 1.5 month. We should not forget to match time units.

$$P\{X > 1.5\} = 1 - F(1.5) = 1 - \left(1 - e^{-4\times 1.5}\right) = e^{-6} \approx 0.0025$$

Exponential RV: Example V

Q7. The mean time between successive telephone calls arriving randomly at a switchboard is 2 minutes. We know that the time between successive telephone calls are distributed exponentially. What is the probability that the time between successive telephone calls will be between 90 and 150 seconds?

A7. Let us construct our random variable in the unit of minutes.

$$E[X] = 2 \text{ min.} \Rightarrow X \sim Exp\left(\lambda = \frac{1}{2}\frac{\text{calls}}{\text{minute}}\right)$$

However, the question asks the probability of an event that is described with seconds. Thus, we are interested in the following probability:

$$P\{1.5 < X < 2.5\}$$

Exponential RV: Example V (cont'd)

A7. Either integrate the PDF:

$$P\{1.5 < X < 2.5\} = \int_{1.5}^{2.5} \frac{1}{2} e^{-\frac{x}{2}} = -e^{-\frac{x}{2}}\Big|_{1.5}^{2.5} = e^{-\frac{3}{4}} - e^{-\frac{5}{4}} \approx 0.1859$$

or use directly the CDF:

$$P\{1.5 < X < 2.5\} = F(2.5) - F(1.5) = \left(1 - e^{-\frac{2.5}{2}}\right) - \left(1 - e^{-\frac{1.5}{2}}\right) \approx 0.1859$$

What is Memoryless Property?

- A random variable X is said to be without memory, or **memoryless,** if:

$$P\{X > s+t | X > t\} = P\{X > s\}, \quad \forall s,t \geq 0$$

- Suppose X denotes the lifetime of some instrument and X is memoryless.
- Then the probability that the instrument lives for at least $s+t$ hours given that it has survived t hours is the same as the initial probability that it lives for at least s hours.
- If the instrument is alive at time t, then the distribution of the remaining amount of time that it survives is the same as the original lifetime distribution.
- The instrument **does not remember** that it has already been in use for a time t.

Equivalent Expressions of Memoryless Property

- A random variable X is said to be without memory, or memoryless, if:

$$P\{X > s+t | X > t\} = P\{X > s\}, \quad \forall s,t \geq 0$$

- The above equation can be expressed by expanding conditional probability:

$$\frac{P\{X > s+t \cap X > t\}}{P\{X > t\}} = P\{X > s\}, \quad \forall s,t \geq 0$$

and since $(X > s+t) \cap (X > t) = X > s+t$, we can write:

$$P\{X > s+t\} = P\{X > t\} P\{X > s\}, \quad \forall s,t \geq 0$$

Exponential RV Has Memoryless Property

- **Proposition:** Exponential RV has memoryless property:
- **Proof:** We can show that the memoryless property holds for exponential RV.

$$\begin{aligned} P\{X > s+t | X > t\} &= \frac{P\{X>s+t \cap X>t\}}{P\{X>t\}} = \frac{P\{X>s+t\}}{P\{X>t\}} \\ &= \frac{1-F(s+t)}{1-F(t)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda t}} \\ &= e^{-\lambda s} = 1 - F(s) = P\{X > s\} \end{aligned}$$

  The above statement is true for all $s, t \geq 0$.
- Or simply put:

$$P\{X > s+t\} = e^{-\lambda(s+t)} = e^{-\lambda s} e^{-\lambda t} = P\{X > t\} P\{X > s\}, \quad \forall s, t \geq 0$$

- Exponential RV is the only continuous RV that has the memoryless property.
- Is there any discrete RV that has the memoryless property?

Exponential RV: Example VI

Q8. The lifetime of a radio is exponentially distributed with a mean of 2.5 years.

a. If Dwight buys a brand new radio, what is the probability that it will be working at least for ten years?

b. If Dwight buys a five-year-old radio, what is the probability that it will be working after an additional ten years?

A8. Let X denote the lifetime of the radio:

$$E[X] = \frac{1}{\lambda} = 2.5 \text{ years} \Rightarrow X \sim Exp\left(\lambda = \frac{1}{2.5} = 0.4 \ \frac{\text{radio}}{\text{year}}\right)$$

Exponential RV: Example VI (cont'd)

a. $P\{X > 10\} = 1 - F(10) = 1 - \left(1 - e^{-0.4\times 10}\right) = e^{-4} \approx 0.0183$

b. Given that observation that the radio has worked for 5 years, that is $X > 5$, we are interested in the prob. that it will work for an additional 10 years, namely the prob. of $X > 15$.

$$\begin{aligned} P\{X > 15 | X > 5\} &= \frac{P\{X>15 \cap X>5\}}{P\{X>5\}} = \frac{P\{X>15\}}{P\{X>5\}} \\ &= \frac{1-F(15)}{1-F(5)} = \frac{1-\left(1-e^{-0.4\times 15}\right)}{1-\left(1-e^{-0.4\times 5}\right)} = \frac{e^{-6}}{e^{-2}} \\ &= e^{-6-(-2)} = e^{-4} = 0.0183 \end{aligned}$$

Or simply put, since exponential RVs are memoryless, additional lifetime of a used radio has same probabilistic properties as the lifetime of a brand new radio, i.e., radio does not wear out. Thus the answer is same as in part (a).

Exponential RV: Example VII

Q9. If the life of a component has an exponential distribution with a mean of 100 hours, what is the probability that it will last 400 hours given that it has already survived 200 hours?

Exponential RV: Example VII

Q9. If the life of a component has an exponential distribution with a mean of 100 hours, what is the probability that it will last 400 hours given that it has already survived 200 hours?

A9. Let X denote the life of the component:

$$E[X] = \frac{1}{\lambda} = 100 \text{ hours} \Rightarrow X \sim Exp\left(\lambda = 0.01 \ \frac{\text{component}}{\text{hour}}\right)$$

The probability of interest is:

$$P\{X > 400 | X > 200\}$$

Since, exponential RV has no memory:

$$\begin{aligned} P\{X > 400 | X > 200\} &= P\{X > 200\} = 1 - F(200) \\ &= 1 - \left(1 - e^{-0.01 \times 200}\right) = e^{-2} \approx 0.1353 \end{aligned}$$

The Origin of Normal Random Variable

- The normal random variable and its distribution have great essence in our further probability and statistical studies.
- We should spare a little more time to understand how this random variable and the corresponding distribution behaves.
- We are going to learn more about this random variable even after this chapter is complete.
- The normal RV is also called a Gaussian RV, named after the mathematician Carl Friedrich Gauss[2], due to his significant contributions on its theory.

[2] Carl Friedrich Gauss, 1777-1855.

Normal RV: PDF

- We say that X is a **normal RV,** or simply that X is **normally distributed,** if its PDF is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

  for $\mu \in \mathbb{R}$ and $\sigma > 0$.
- Here:
  - μ is called the **location parameter,**
  - σ is called the **scale parameter.**

Normal RV: PDF (cont'd)

- For different values of $\mu \in \mathbb{R}$ and $\sigma > 0$, we get a different normal random variable.

$$X \sim N(\mu, \sigma)$$

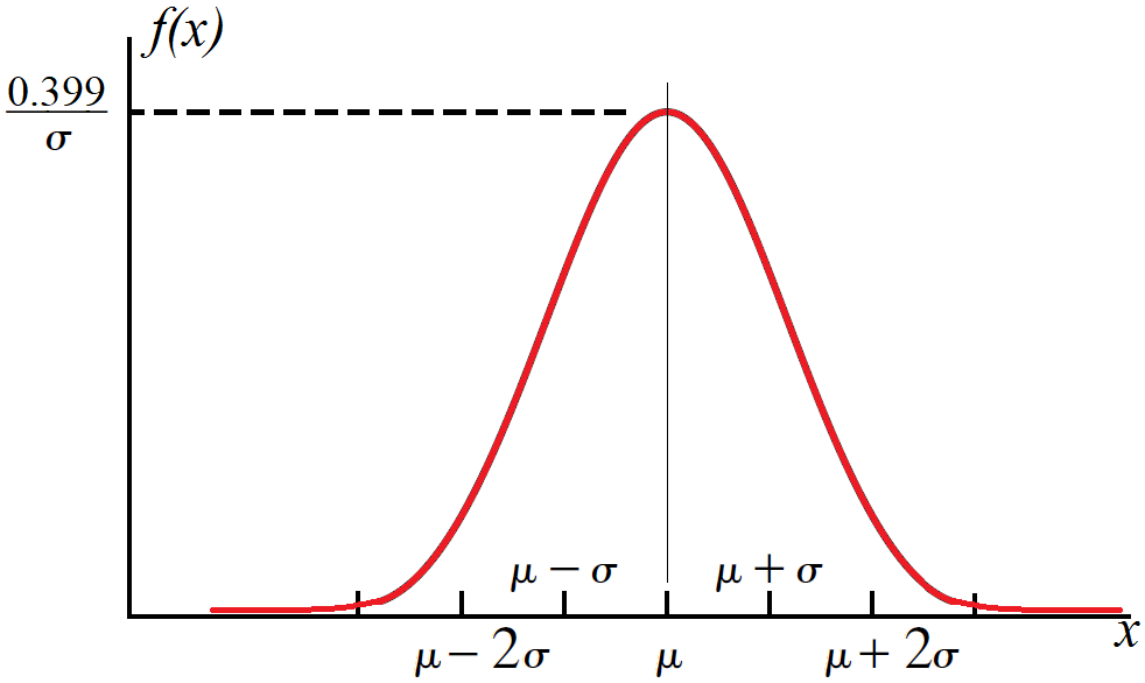
- In all cases, the area under the PDF is always one[3].

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\,dx = 1$$

[3] See page 198-199 of Ross's textbook.

Normal RV: PDF (cont'd)

- The PDF of a normal RV is a bell-shaped curve that is symmetric around the location parameter μ.
- The function converges to zero at both extremes.



Standard Normal RV: PDF

- When the location and the scale parameters of the normal RV are chosen as $\mu = 0$ and $\sigma = 1$, the resulting normal random variable is called the "standard normal RV".
- The standard normal variable is usually denoted by Z.

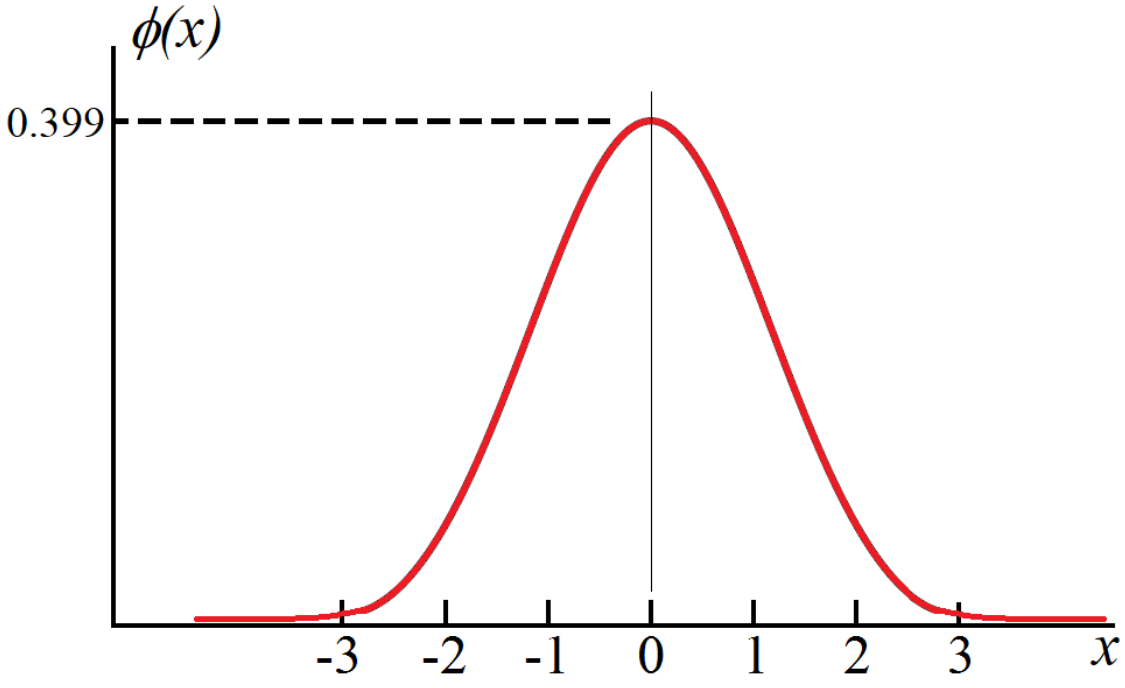
$$Z \sim N(0,1) \sim N(\mu = 0, \sigma = 1)$$

- The PDF of the standard normal RV can be computed by plugging $\mu = 0$ and $\sigma = 1$ in the PDF of the normal RV, and be denoted by ϕ:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty$$

Standard Normal RV: PDF (cont'd)

- The PDF of the standard normal RV is symmetric around the location parameter, which is $\mu = 0$.
- The function converges to zero at both extremes.



Normal RV: CDF

- The cumulative distribution function (CDF) of the normal RV is calculated as:

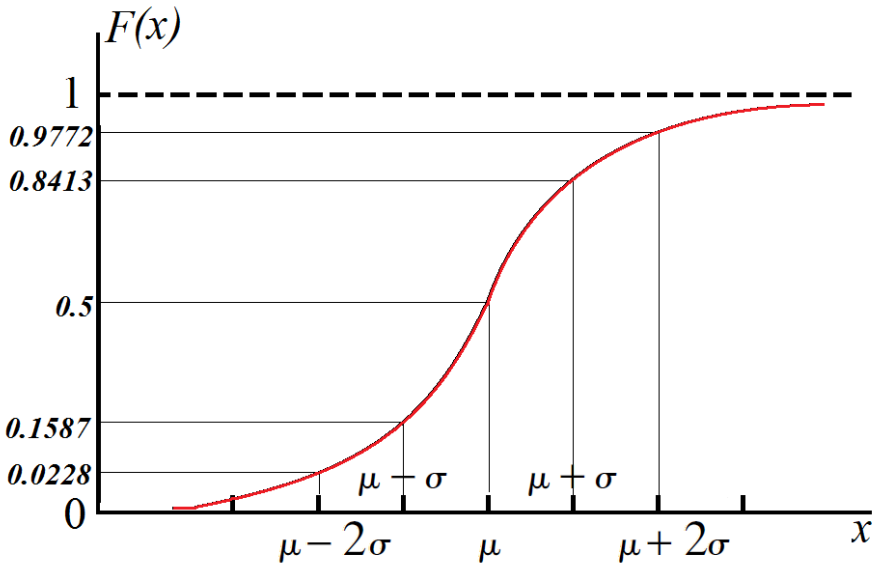
$$F(x) = P\{X \leq x\} = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

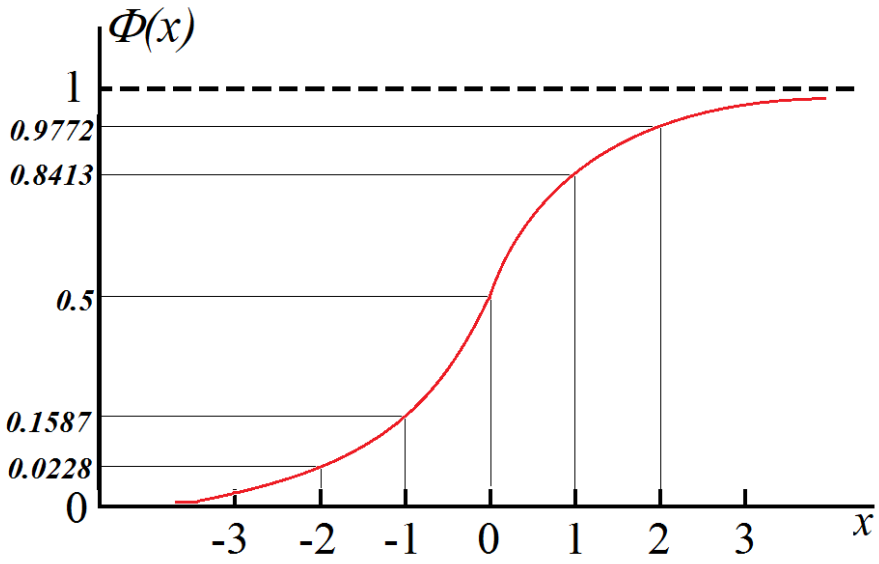
- However, the above integral is not solvable.
- The CDF of the normal random variable is not available in closed-form (but available numerically).
- We will use:
  - F to denote the CDF of the normal random variable and
  - Φ to denote the CDF of the standard normal variable, that is:

$$\Phi(x) = P\{Z \leq x\} = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

Normal RV: CDF (cont'd)

- The CDF of a normal RV is an increasing S-shaped curve.
  - At $-\infty$, it converges to 0.
  - At ∞, it converges to 1.





Linear Transformation of Normal RVs: Proposition I

- **Proposition I:** If X is normally distributed with location parameter μ and scale parameter σ

$$X \sim N(\mu, \sigma)$$

then, for $a, b \in \mathbb{R}$, $Y = aX + b$ is normally distributed with location parameter $a\mu + b$ and scale parameter $|a\sigma|$:

$$Y = aX + b \sim N(a\mu + b, |a\sigma|)$$

Linear Transformation of Normal RVs: Proposition I (cont'd)

- **Proof:** Suppose[4] that $a > 0$. Let F_Y denote the CDF of Y. Then:

$$F_Y(x) = P\{Y \leq x\} = P\{aX + b \leq x\} = P\left\{X \leq \frac{x-b}{a}\right\} = F_X\left(\frac{x-b}{a}\right)$$

where F_X is the CDF of X. By differentiation, the PDF of Y can be computed:

$$\begin{aligned} f_Y(x) &= \frac{\partial F_Y(x)}{\partial x} = \frac{\partial F_X\left(\frac{x-b}{a}\right)}{\partial x} \\ &= \frac{\partial F_X\left(\frac{x-b}{a}\right)}{\partial x} = \frac{1}{a} f_X\left(\frac{x-b}{a}\right) \\ &= \frac{1}{a\sigma\sqrt{2\pi}} \exp\left\{-\frac{\left(\frac{x-b}{a}-\mu\right)^2}{2\sigma^2}\right\} \\ &= \frac{1}{a\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-(a\mu+b))^2}{2(a\sigma)^2}\right\} \end{aligned}$$

The PDF of Y shows that we have $a\mu + b$ for the location and $a\sigma$ for the scale parameter.

[4] The proof is similar when $a < 0$ and trivial when $a = 0$.

Linear Transformation of Normal RVs: Proposition II

- **Proposition II:** If X is normally distributed with location parameter μ and scale parameter σ

$$X \sim N(\mu, \sigma)$$

then, $Z = \frac{X-\mu}{\sigma}$ is a standard normal RV.

- **Proof:** The above transformation is a linear transformation, in which $a = \frac{1}{\sigma}$ and $b = -\frac{\mu}{\sigma}$. Then, by the previous proposition, the transformed variable is also normal with the parameters of standard normal RV:
  - The location parameter $\frac{1}{\sigma}\mu + \left(-\frac{\mu}{\sigma}\right) = 0$.
  - The scale parameter $\left(\frac{1}{\sigma}\right)\sigma = 1$.

Linear Transformation of Normal Random Variables

The results of propositions I and II:

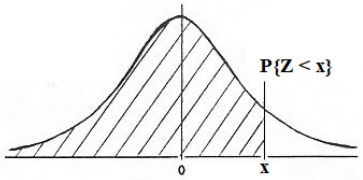
- Normal random variables do not lose their probabilistic properties under linear transformation.
- For random variable $X \sim N(\mu, \sigma)$, if one is interested in calculating $F(x) = P\{X \leq x\}$, we can apply the following transformation:

$$F(x) = P\{X \leq x\} = P\left\{\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right\} = P\left\{Z \leq \frac{x-\mu}{\sigma}\right\} = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

- If we can store approximate CDF values for a standard normal RV in a table, then we can compute probabilities regarding to all normal RVs.

The CDF Table For Standard Normal Random Variable

- The table shows $\Phi(x) = P\{Z \leq x\}$ for many $x \geq 0$.
- We can also derive $\Phi(x)$ for $x < 0$, since the PDF is symmetric around zero.
- If x does not exist in the table, we can approximate $\Phi(x)$ by **linear interpolation** between the closest values.
- We can also derive the inverse function $\Phi^{-1}(p)$ for probabilities $p \in (0, 1)$.



x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
x	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Standard Normal RV: Example

Q10. Consider Z as a standard normal random variable.

$$Z \sim N(0,1)$$

Calculate the following probabilities using the CDF table of standard normal variable.

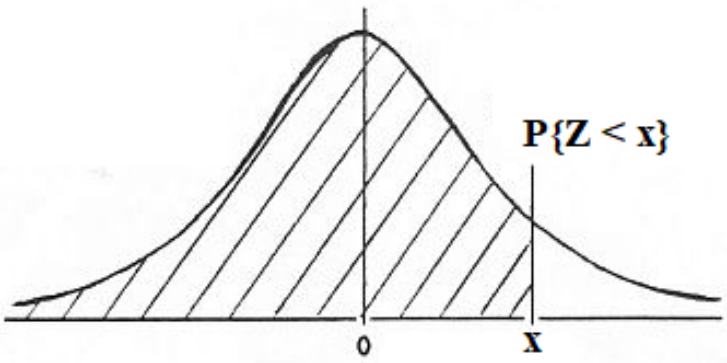
a. $P\{Z \leq 0.34\}$
b. $P\{Z > 1.667\}$
c. $P\{Z \geq -1.08\}$
d. $P\{Z < -3.20\}$
e. $P\{-0.98 \leq Z \leq 0.49\}$

A10. Note that, since normal RV is a continuous RV, there is no difference between $P\{Z \leq x\}$ and $P\{Z < x\}$. All the answers can be derived from the "CDF Table For Standard Normal RV".

Standard Normal RV: Example (cont'd)

a. $P\{Z \leq 0.34\}$ can directly be derived from the table as:

$$P\{Z \leq 0.34\} = \Phi(0.34) \approx 0.6331$$



x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Standard Normal RV: Example (cont'd)

b. $P\{Z > 1.667\} = 1 - P\{Z \leq 1.667\} = 1 - \Phi(1.667)$
However, $\Phi(1.667)$ is not directly available in the table. Hence, we apply a linear approximation.

$$\begin{aligned}\Phi(1.667) &\approx 0.3 \times \Phi(1.66) + 0.7 \times \Phi(1.67) \\ &= 0.3 \times 0.9515 + 0.7 \times 0.9525 = 0.9522\end{aligned}$$

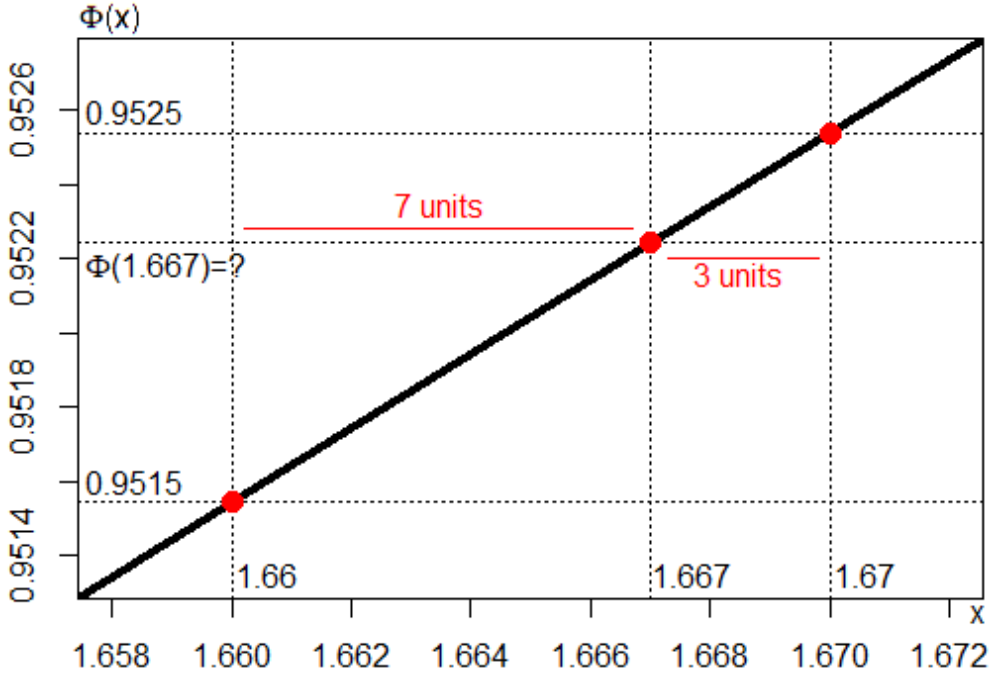
The answer is:

$$P\{Z > 1.667\} \approx 1 - 0.9522 = 0.0478$$

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Standard Normal RV: Example II (cont'd)

The following figure explains how linear approximation works.

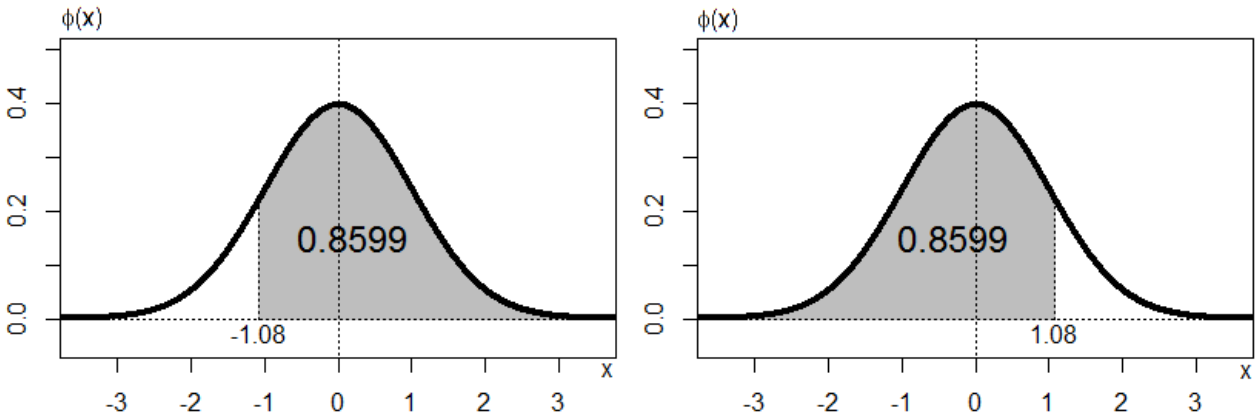


$$\Phi(1.667) \approx \frac{3\times\Phi(1.66)+7\times\Phi(1.67)}{10} \approx \frac{3\times 0.9515+7\times 0.9525}{10} \approx 0.9522$$

Standard Normal RV: Example (cont'd)

c. In order to find $P\{Z \geq -1.08\}$, we will make use of the symmetricity of the standard normal PDF around zero:

$$P\{Z \geq -1.08\} = P\{Z \leq 1.08\} = \Phi(1.08)$$



The table, then, tells us $\Phi(1.08) \approx 0.8599$.

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.0	~~0.8413~~	~~0.8438~~	~~0.8461~~	~~0.8485~~	~~0.8508~~	~~0.8531~~	~~0.8554~~	~~0.8577~~	0.8599	0.8621
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Standard Normal RV: Example (cont'd)

d. For $P\{Z < -3.20\}$ as well, we will make use of the symmetricity property:

$$\begin{aligned} P\{Z < -3.20\} &= P\{Z > 3.20\} = 1 - P\{Z \leq 3.20\} \\ &= 1 - \Phi(3.20) \approx 1 - 0.9993 = 0.0007 \end{aligned}$$

where $\Phi(3.20) \approx 0.9993$ is derived from the table (at the bottom).

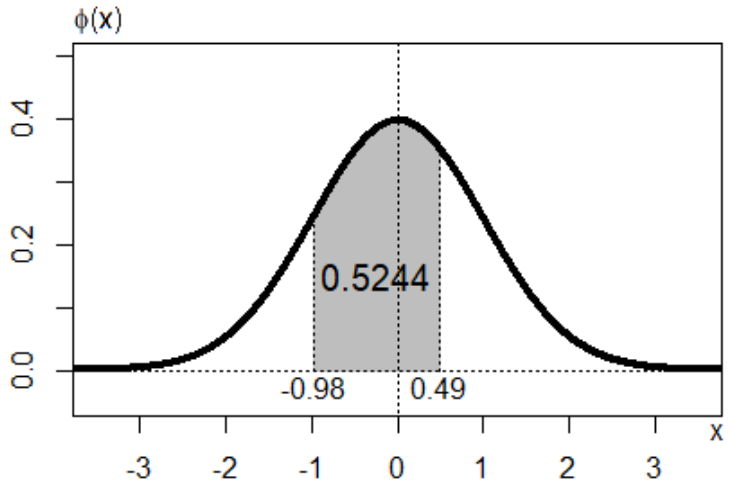
x	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Standard Normal RV: Example (cont'd)

e. For $P\{-0.98 \leq Z < 0.49\}$, we need to write:

$$\begin{aligned} P\{-0.98 \leq Z \leq 0.49\} &= P\{Z \leq 0.49\} - P\{Z < -0.98\} \\ &= P\{Z \leq 0.49\} - P\{Z > 0.98\} \\ &= P\{Z \leq 0.49\} - (1 - P\{Z \leq 0.98\}) \\ &= \Phi(0.49) - (1 - \Phi(0.98)) \\ &\approx 0.6879 - (1 - 0.8365) = 0.5244 \end{aligned}$$

App. values of $\Phi(0.49)$ and $\Phi(0.98)$ are derived from the table.



Standard Normal RV: Example II

Q11. Consider Z as a standard normal random variable. $Z \sim N(0,1)$

a. Identify x for which $P\{Z \leq x\} = 0.60$.
b. Identify x for which $P\{Z \leq x\} = 0.10$.
c. Identify x for which $P\{-x \leq Z \leq x\} = 0.95$.

A11. All these question corresponds to the inverse CDF of the standard normal RV. Therefore, we will search the "CDF Table For Standard Normal RV" through the reverted perspective.

Standard Normal RV: Example II (cont'd)

a. $P\{Z \leq x\} = 0.60 \Rightarrow \Phi(x) = 0.60 \Rightarrow \Phi^{-1}(0.60) = x$ Unfortunately, in the contents of the table, there is not an exact probability of 0.60, but close values:

$$\Phi^{-1}(0.5987) \approx 0.25, \quad \Phi^{-1}(0.6026) \approx 0.26$$

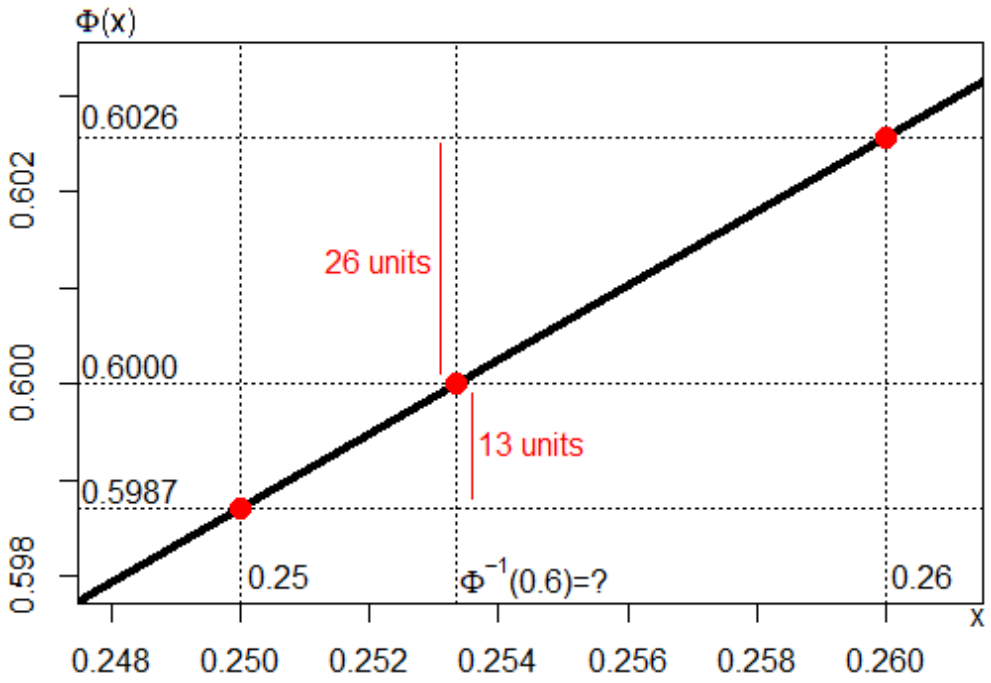
x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

We can approximate an intermediate value linearly.

$$\Phi^{-1}(0.60) \approx \frac{26 \times \Phi^{-1}(0.5987) + 13 \times \Phi^{-1}(0.6026)}{39} \approx 0.2533$$

Standard Normal RV: Example II (cont'd)

The following figure explains how linear approximation works.

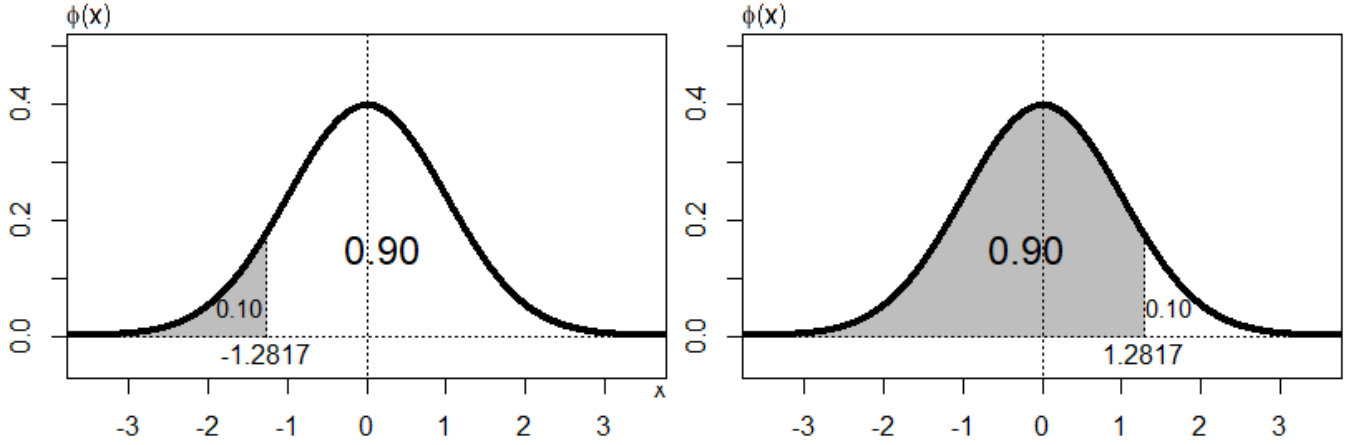


$$\Phi^{-1}(0.60) \approx \frac{26\times\Phi^{-1}(0.5987)+13\times\Phi^{-1}(0.6026)}{39} \approx \frac{26\times 0.25+13\times 0.26}{39} \approx 0.2533$$

Standard Normal RV: Example II (cont'd)

b. $P\{Z \leq x\} = 0.10 \Rightarrow \Phi(x) = 0.10 \Rightarrow \Phi^{-1}(0.10) = x$
Note that, the table provides cumulative probabilities which are greater than or equal to 0.5. However, we can again make use of the symmetricity property. Notice that:

$$\Phi^{-1}(0.10) = -\Phi^{-1}(1-0.10) = -\Phi^{-1}(0.90) \approx -1.2817$$



$\Phi^{-1}(0.90)$ is approximated with the same manner as in part (a).

Standard Normal RV: Example II (cont'd)

c. In order to identify x for which $P\{-x \leq Z \leq x\} = 0.95$, we construct:

$$P\{-x \leq Z \leq x\} = \Phi(x) - \Phi(-x) = \Phi(x) - [1 - \Phi(x)]$$

using the symmetricity property. Thus:

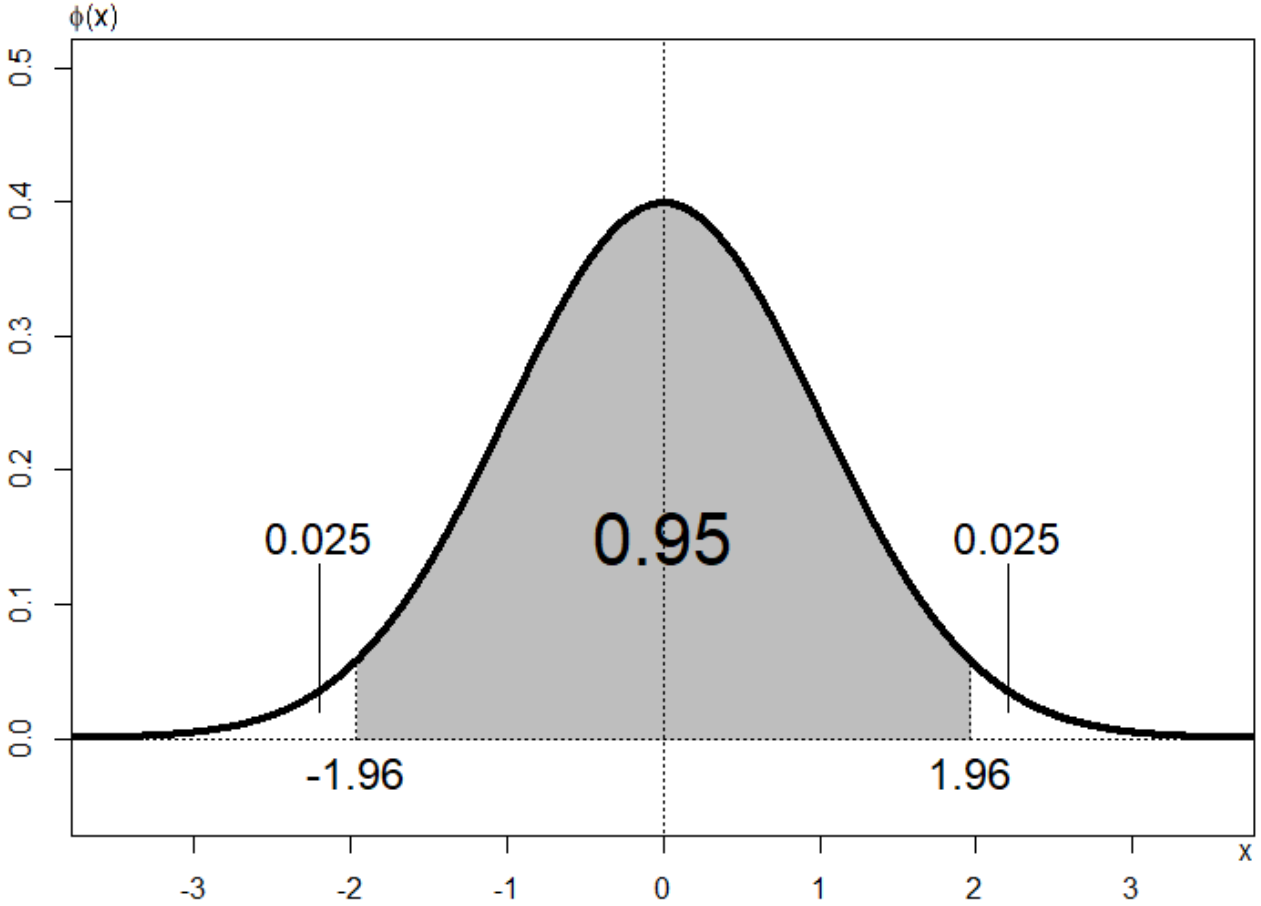
$$\Phi(x) - [1 - \Phi(x)] = 0.95 \quad \Rightarrow \Phi(x) = \frac{0.95+1}{2} = 0.975$$
$$\Rightarrow \Phi^{-1}(0.975) = x$$

all we need to do is to identify x for which $P\{Z \leq x\} = 0.975$

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

$$\Phi^{-1}(0.975) \approx 1.96$$

Standard Normal RV: Example II (cont'd)



Normal RV: Example

Q12. Calculate the following probabilities for non-standard random variables.

a. For $X \sim N(\mu = -3, \sigma = 1)$, calculate $P\{X < -3.75\}$
b. For $X \sim N(\mu = 5, \sigma = 1.2)$, calculate $P\{X > 4\}$
c. For $X \sim N(\mu = 35, \sigma = 15)$, calculate $P\{X > 52\}$
d. For $X \sim N\left(\mu = 4, \sigma = \sqrt{20}\right)$, calculate $P\{0 < X < 3\}$

A12. All of the above probability statements can be linearly transformed[5] by

$$Z = \frac{X-\mu}{\sigma}$$

so that they correspond to a statement about standard normal RV. Then, we can search probabilities through the "CDF Table For Standard Normal RV".

[5] Inequalities are not affected by linear transformations.

Normal RV: Example (cont'd)

a. $X \sim N(\mu = -3, \sigma = 1) \Rightarrow P\{X < -3.75\}$

$$\begin{aligned} P\{X < -3.75\} &= P\left\{\frac{X-\mu}{\sigma} < \frac{-3.75-\mu}{\sigma}\right\} = P\left\{Z < \frac{-3.75-(-3)}{1}\right\} \\ &= P\{Z < -0.75\} = \Phi(-0.75) \\ &= 1 - \Phi(0.75) \approx 1 - 0.7734 = 0.2266 \end{aligned}$$

b. $X \sim N(\mu = 5, \sigma = 1.2) \Rightarrow P\{X > 4\}$

$$\begin{aligned} P\{X > 4\} &= P\left\{\frac{X-\mu}{\sigma} > \frac{4-\mu}{\sigma}\right\} = P\left\{Z > \frac{4-5}{1.2}\right\} \\ &= P\left\{Z > -\frac{5}{6}\right\} = P\left\{Z < \frac{5}{6}\right\} = \Phi\left(\frac{5}{6}\right) \\ &\approx \frac{2\times\Phi(0.83)+\Phi(0.84)}{3} \approx \frac{2\times 0.7967+0.7995}{3} \approx 0.7976 \end{aligned}$$

Normal RV: Example (cont'd)

c. $X \sim N(\mu = 35, \sigma = 15) \Rightarrow P\{X > 52\}$

$$\begin{aligned} P\{X > 52\} &= P\left\{\frac{X-\mu}{\sigma} > \frac{52-\mu}{\sigma}\right\} = P\left\{Z > \frac{52-35}{15}\right\} \\ &= P\left\{Z > \frac{17}{15}\right\} = 1 - P\left\{Z < \frac{17}{15}\right\} = 1 - \Phi\left(\frac{17}{15}\right) \\ &\approx 1 - \frac{2\times\Phi(1.13)+\Phi(1.14)}{3} \approx 1 - \frac{2\times 0.8708+0.8729}{3} \\ &= 1 - 0.8715 = 0.1285 \end{aligned}$$

Normal RV: Example (cont'd)

d. $X \sim N\left(\mu = 4, \sigma = \sqrt{20}\right) \Rightarrow P\{0 < X < 3\}$

$$
\begin{aligned}
P\{0 < X < 3\} &= P\left\{\frac{0-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{3-\mu}{\sigma}\right\} \\
&= P\left\{\frac{0-4}{\sqrt{20}} < Z < \frac{3-4}{\sqrt{20}}\right\} \\
&\approx P\{-0.894 < Z < -0.224\} \\
&= \Phi(-0.224) - \Phi(-0.894) \\
&= (1 - \Phi(0.224)) - (1 - \Phi(0.894)) \\
&= \Phi(0.894) - \Phi(0.224) \\
&\approx \left(\frac{6\times\Phi(0.89)+4\times\Phi(0.90)}{10}\right) - \left(\frac{6\times\Phi(0.22)+4\times\Phi(0.23)}{10}\right) \\
&\approx \left(\frac{6\times 0.8133+4\times 0.8159}{10}\right) - \left(\frac{6\times 0.5871+4\times 0.5910}{10}\right) \\
&\approx 0.8143 - 0.5886 = 0.2257
\end{aligned}
$$

Normal RV: Example II

Q13. We consider the percent purity of oxygen from a certain supplier. Assume that the distribution of percent purity was approximately normal with $\mu = 99.61$ and $\sigma = 0.08$.

a. What percentage of purity values would you expect to be between 99.6 and 99.7?
b. What percentage of purity values is less than 0.1 away from μ?
c. What purity value would you expect to exceed exactly 7% of the population?
d. What purity value is exceeded by exactly 3% of the population?

A13. While parts (a) and (b) can be solved by writing the probability statements and, then, through linear transformation, parts (c) and (d) requires the reverted perspective.

Normal RV: Example II (cont'd)

a. $X \sim N(\mu = 99.61, \sigma = 0.08) \Rightarrow P\{99.6 \leq X \leq 99.7\}$

$$
\begin{aligned}
P\{99.6 \leq X \leq 99.7\} &= P\left\{\tfrac{99.6-99.61}{0.08} \leq Z \leq \tfrac{99.7-99.61}{0.08}\right\} \\
&= P\{-0.125 \leq Z \leq 1.125\} \\
&= \Phi(1.125) - \Phi(-0.125) \\
&= \Phi(1.125) - [1 - \Phi(0.125)] \\
&\approx \tfrac{\Phi(1.12)+\Phi(1.13)}{2} - \left[1 - \tfrac{\Phi(0.12)+\Phi(0.13)}{2}\right] \\
&\approx \tfrac{0.8686+0.8708}{2} - \left[1 - \tfrac{0.5478+0.5517}{2}\right] \\
&\approx 0.8697 - [1 - 0.5498] = 0.4195
\end{aligned}
$$

Normal RV: Example II (cont'd)

b. $X \sim N(\mu = 99.61, \sigma = 0.08) \Rightarrow P\{99.51 \leq X \leq 99.71\}$

$$
\begin{aligned}
P\{99.51 < X < 99.71\} &= P\left\{\frac{99.51-99.61}{0.08} < Z < \frac{99.71-99.61}{0.08}\right\} \\
&= P\{-1.25 \leq Z \leq 1.25\} \\
&= \Phi(1.25) - \Phi(-1.25) \\
&= \Phi(1.25) - [1 - \Phi(1.25)] \\
&= 2 \times \Phi(1.25) - 1 \\
&\approx 2 \times 0.8944 - 1 = 0.7888
\end{aligned}
$$

Normal RV: Example II (cont'd)

c. $X \sim N(\mu = 99.61, \sigma = 0.08) \Rightarrow P\{X \leq x\} = 0.07$

$$P\{X \leq x\} = P\left\{Z \leq \frac{x - 99.61}{0.08}\right\} = \Phi\left(\frac{x - 99.61}{0.08}\right) = 0.07$$

$$\Phi^{-1}(0.07) = \frac{x - 99.61}{0.08}$$

$$\Phi^{-1}(0.93) \approx 1.476 \Rightarrow \Phi^{-1}(0.07) \approx -1.476$$

$$\frac{x - 99.61}{0.08} = -1.476 \Rightarrow x \approx 99.492$$

Normal RV: Example II (cont'd)

d. $X \sim N(\mu = 99.61, \sigma = 0.08) \Rightarrow P\{X > x\} = 0.03$

$$P\{X > x\} = 0.03 \Rightarrow P\{X \leq x\} = 0.97$$

$$P\{X \leq x\} = P\left\{Z \leq \frac{x - 99.61}{0.08}\right\} = \Phi\left(\frac{x - 99.61}{0.08}\right) = 0.97$$

$$\Phi^{-1}(0.97) = \frac{x - 99.61}{0.08} \approx 1.881 \Rightarrow x \approx 99.7605$$

Normal RV: Example III

Q14. James receives a package everyday at work. In 25% of the days, packages are classified as type A, and on remaining days, packages are classified as type B. The weight (in grams) of a:

- type A package is $N(\mu = 100, \sigma = 2)$
- type B package is $N(\mu = 103, \sigma = 3)$

On any day, what is the probability that the recieved package weighs less than or equal to 98.5 grams?

A14. Let:

- A: the event of receiving type A package, $P(A) = 0.25$
- B: the event of receiving type B package, $P(B) = 0.75$
- X: the weight of the received package.

We do not know the distribution of X, but we know its conditional distributions.

$$X|A \sim X_A \sim N(\mu = 100, \sigma = 2), \quad X|B \sim X_B \sim N(\mu = 103, \sigma = 3)$$

Normal RV: Example III (cont'd)

A14. The question asks $P\{X \leq 98.5\}$

$$
\begin{aligned}
P\{X \leq 98.5\} &= P(A)P\{X \leq 98.5|A\} + P(B)P\{X \leq 98.5|B\} \\
&= P(A)P\{X_A \leq 98.5\} + P(B)P\{X_B \leq 98.5\} \\
&= P(A)P\left\{Z \leq \tfrac{98.5-100}{2}\right\} + P(B)P\left\{Z \leq \tfrac{98.5-103}{3}\right\} \\
&= P(A)P\{Z \leq -0.75\} + P(B)P\{Z \leq -1.5\} \\
&= P(A)\Phi(-0.75) + P(B)\Phi(-1.5) \\
&\approx 0.25 \times 0.2266 + 0.75 \times 0.0668 \\
&\approx 0.1068
\end{aligned}
$$

Normal RV: Example IV

Q15. The results of a statistics exam are assumed to be normally distributed with $\mu = 65$ and $\sigma = 15$. In order to get an AA, a student must get a grade greater than at least 90% of the students. Similarly, in order to get a BA, the student must get a grade greater than at least 80% of the students.

a. What is the minimum score required to get an AA?
b. What is the minimum score required to get at least BA?

A15. A random student's grade is normally distributed:

$$X \sim N(\mu = 65, \sigma = 15)$$

Let x and y be the minimum scores required to get AA and BA, respectively. Then, we have:

- $P\{X \leq x\} = 0.90$ for part (a).
- $P\{X \leq y\} = 0.80$ for part (b).

Normal RV: Example IV (cont'd)

a. $\Phi^{-1}(0.90) \approx 1.2817$ is already computed in Question 11.

$$P\{X \leq x\} = P\left\{\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right\} = P\left\{Z \leq \frac{x-65}{15}\right\} = \Phi\left(\frac{x-65}{15}\right) = 0.90$$

$$\Phi^{-1}(0.90) = \frac{x-65}{15} \approx 1.2817 \Rightarrow x \approx 84.23$$

b. $\Phi^{-1}(0.80) \approx 0.8418$ is computed through linear approximation.

$$P\{X \leq y\} = P\left\{\frac{X-\mu}{\sigma} \leq \frac{y-\mu}{\sigma}\right\} = P\left\{Z \leq \frac{y-65}{15}\right\} = \Phi\left(\frac{y-65}{15}\right) = 0.80$$

$$\Phi^{-1}(0.7995) = 0.84, \quad \Phi^{-1}(0.8023) = 0.85$$

$$\Phi^{-1}(0.80) = \frac{y-65}{15} \approx \frac{23 \times 0.84 + 5 \times 0.85}{28} \approx 0.8418 \Rightarrow y \approx 77.63$$

Normal RV: Expected Value

- We, first, compute the expected value of a standard normal RV.
- Notice the following equations regarding to PDF of standard normal variable:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$\begin{aligned}
\phi'(x) &= \frac{-x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} = -x\phi(x) && \Rightarrow x\phi(x) = -\phi'(x) \\
\phi''(x) &= (-1)\phi(x) + (-x\phi(x))(-x) \\
&= \left(x^2 - 1\right)\phi(x) && \Rightarrow x^2\phi(x) = \phi''(x) + \phi(x)
\end{aligned}$$

Normal RV: Expected Value (cont'd)

- For $Z \sim N(0,1)$, the expected value is:

$$E[Z] = \int_{-\infty}^{\infty} x\phi(x)\,dx = -\int_{-\infty}^{\infty} \phi'(x)\,dx = -[\phi(x)]_{-\infty}^{\infty} = -(0-0) = 0$$

- For[6] any normal RV $X \sim N(\mu, \sigma)$:

$$Z = \frac{X-\mu}{\sigma} \Rightarrow X = \sigma Z + \mu$$

$$E[X] = E[\sigma Z + \mu] = \sigma E[Z] + \mu = \sigma \times 0 + \mu = \mu$$

- The **expected value (mean)** of a normal random variable is equal to its location parameter μ.

[6] See page 72 of "LS04.pdf" file.

Normal RV: Variance

- Now, let us compute the variance of a standard normal RV.
- For $Z \sim N(0,1)$, $E\left[Z^2\right]$ is:

$$\begin{aligned} E\left[Z^2\right] &= \int_{-\infty}^{\infty} x^2 \phi(x)\, dx = \int_{-\infty}^{\infty} \phi''(x) + \phi(x)\, dx \\ &= \int_{-\infty}^{\infty} \phi''(x)\, dx + \int_{-\infty}^{\infty} \phi(x)\, dx = \left[\phi'(x)\right]_{-\infty}^{\infty} + 1 = (0-0)+1 = 1 \end{aligned}$$

$$Var[Z] = E\left[Z^2\right] - E[Z]^2 = 1 - 0^2 = 1$$

- For[7] any normal RV $X \sim N(\mu, \sigma)$:

$$Z = \frac{X-\mu}{\sigma} \Rightarrow X = \sigma Z + \mu$$

$$Var[X] = Var[\sigma Z + \mu] = \sigma^2 Var[Z] = \sigma^2 \times 1 = \sigma^2$$

- The **variance** of a normal random variable is equal to σ^2.

[7] See page 73 of "LS04.pdf" file.

Normal RV: Example V

Q16. The body-weight of male students in Özyeğin University is normally distributed with 78 kg. mean and 25 kg^2 variance.

a. What is the probability of a randomly selected male student to weigh less than 69.75 kg.?

b. What kg. value is exceeded by only a 5% of the male students?

A16. The weight of a random male student is:

$$N(\mu = 78, \sigma = 5)$$

Normal RV: Example V (cont'd)

a. Here, $\Phi(1.65) \approx 0.9505$ is obtained from the "CDF Table for Std. Normal Dist.".

$$P\{X \leq 69.75\} = P\left\{\frac{X-\mu}{\sigma} \leq \frac{69.75-\mu}{\sigma}\right\} = P\left\{Z \leq \frac{69.75-78}{5}\right\} = P\{Z \leq -1.65\}$$

$$P\{Z \leq -1.65\} = \Phi(-1.65) = 1 - \Phi(1.65) \approx 1 - 0.9505 = 0.0495$$

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Normal RV: Example V (cont'd)

b. The question asks for a threshold x such that $P\{X > x\} = 0.05$, or equivalently $P\{X \leq x\} = 0.95$.

$$P\{X \leq x\} = P\left\{\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right\} = P\left\{Z \leq \frac{x-78}{5}\right\} = \Phi\left(\frac{x-78}{5}\right) = 0.95$$

Note that, according to the "CDF Table for Std. Normal Dist.":

$$\Phi^{-1}(0.9495) \approx 1.64, \quad \Phi^{-1}(0.9505) \approx 1.65$$

By a linear approximation, we can compute the threshold x:

$$\Phi^{-1}(0.95) = \frac{x-78}{5} \approx \frac{1.64+1.65}{2} \approx 1.645 \Rightarrow x \approx 86.225$$

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Further Facts on Normal Random Variables

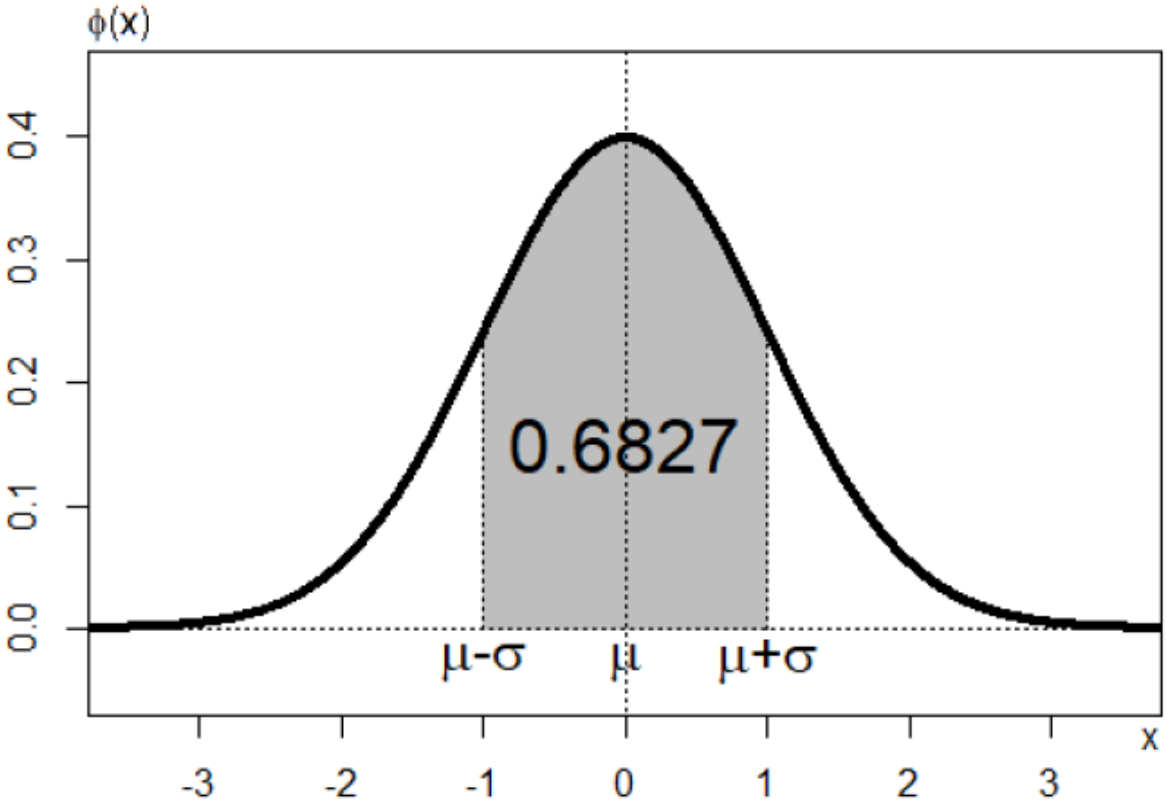
- Say X is a normal RV with location parameter μ and scale parameter σ, or in short

$$X \sim N(\mu, \sigma)$$

- μ is also called the **mean** of the normal distribution. It determines the location of the distribution.
- σ is also called the **standard deviation** of the normal distribution. It determines the dispersion of the distribution.
- σ^2 is called the **variance** of the normal distribution. Though the variance is simply the square of the standard deviation, both terms are essential and they have separate uses in statistics.

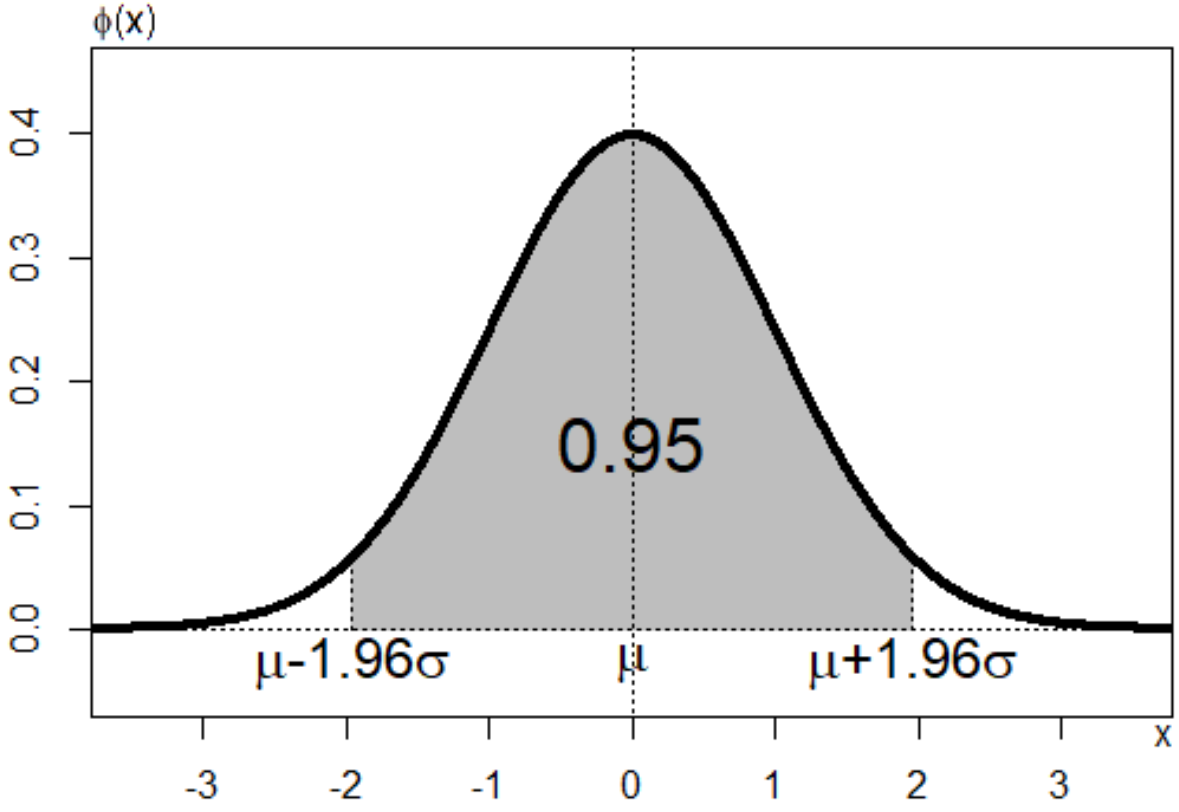
Further Facts on Normal Random Variables (cont'd)

- For any normal RV: $P\{\mu - \sigma \leq X \leq \mu + \sigma\} \approx 0.6827$



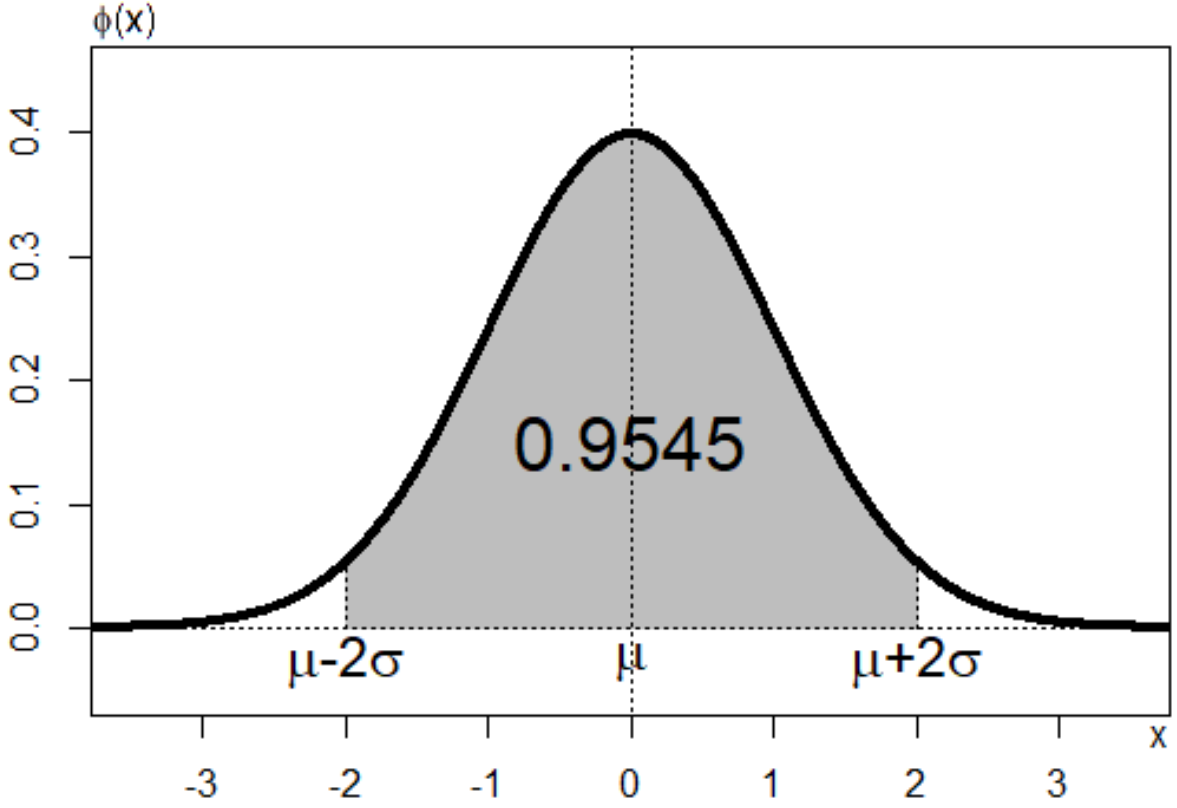
Further Facts on Normal Random Variables (cont'd)

- For any normal RV: $P\{\mu - 1.96\sigma \leq X \leq \mu + 1.96\sigma\} \approx 0.95$



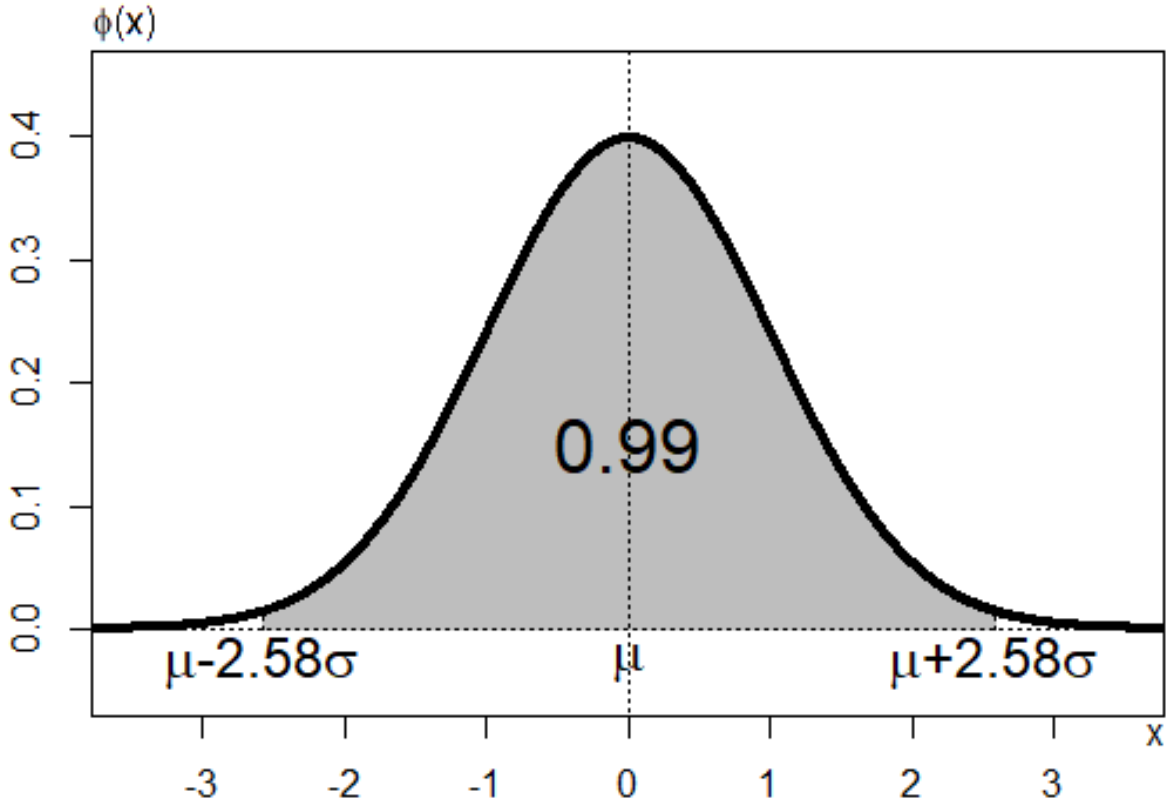
Further Facts on Normal Random Variables (cont'd)

- For any normal RV: $P\{\mu - 2\sigma \leq X \leq \mu + 2\sigma\} \approx 0.9545$



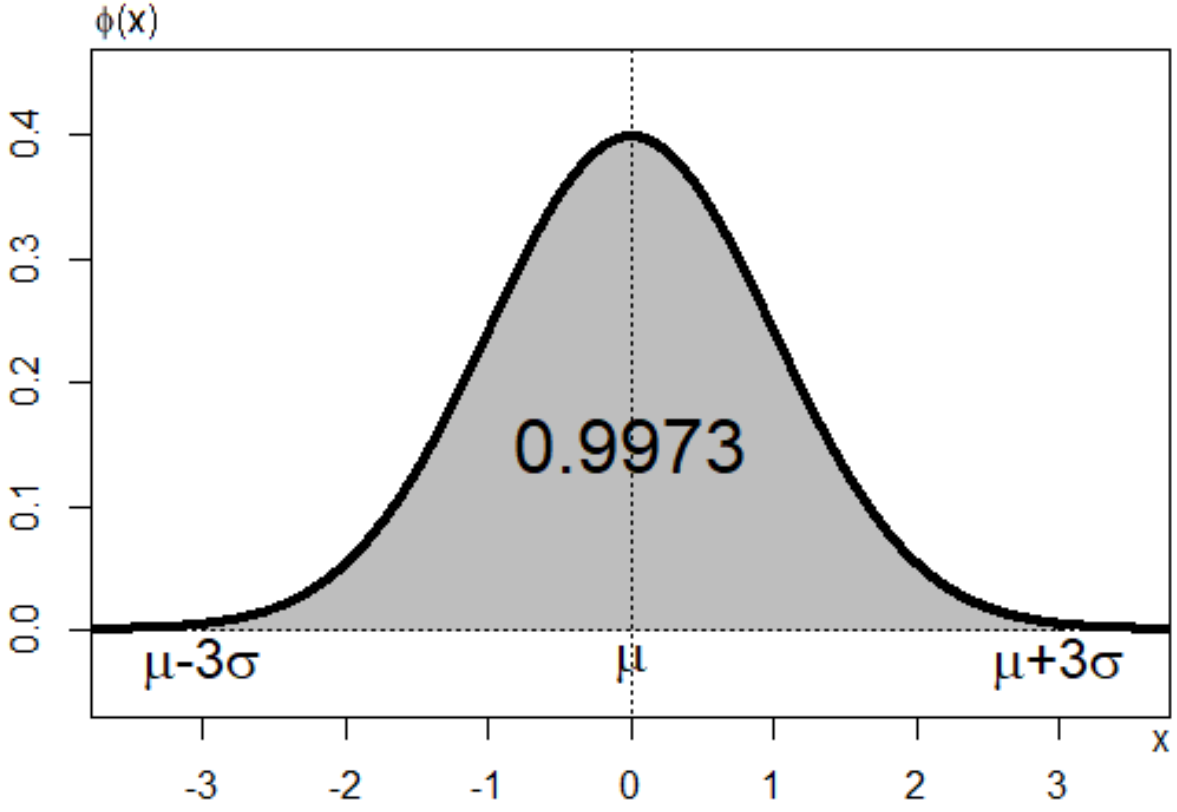
Further Facts on Normal Random Variables (cont'd)

- For any normal RV: $P\{\mu - 2.58\sigma \leq X \leq \mu + 2.58\sigma\} \approx 0.99$



Further Facts on Normal Random Variables (cont'd)

- For any normal RV: $P\{\mu - 3\sigma \leq X \leq \mu + 3\sigma\} \approx 0.9973$



Other Special Continuous Random Variables

- There are other special continuous random variables used for statistical modeling of values in real-life. These are:
  - Triangular Random Variable
  - Lognormal Random Variable
  - Gamma Random Variable
  - Erlang[8] Random Variable
  - Weibull[9] Random Variable
  - Beta Random Variable
  - t-distributed Random Variable
  - Chi-squared Random Variable
  - F-distributed Random Variable

  and many others. Some of which, you will see in MATH 216 and IE 325 courses.

[8] Ernst Hjalmar Waloddi Weibull, 1887-1979.
[9] Agner Krarup Erlang, 1878-1929.

END OF LECTURE SLIDE 06